

opensource.elixpo.com — Platform Plan

A Platform-as-a-Service (PaaS) for hosting, running, and tracking open source initiatives — competitions, hackathons, mentorship programs, and long-running OSS projects — with deep GitHub integration and built-in gamification.

Domain: `opensource.elixpo.com` **Positioning:** "Bring your GitHub org (or repos). We handle the rest — dashboards, mentors, leaderboards, and payouts."

1. Vision

Open source programs today are stitched together with spreadsheets, Discord bots, GitHub Projects, and manual payout tracking. `opensource.elixpo.com` unifies this into a single workspace where an organization can:

- Spin up a competition, program, or initiative in minutes.
- Connect either a **full GitHub organization** or a **curated set of repositories**.
- Track every contributor, mentor, and admin action in real time.
- Gamify participation with points, badges, streaks, and leaderboards.
- Run the entire lifecycle — onboarding → contribution → review → payout — from one dashboard.

Elixpo owns and operates:

- **Account management** (auth, identity, org membership, KYC where required)
- **Transactional & campaign email** (invites, digests, notifications, reminders)
- **Payouts** (prize disbursement, mentor stipends, contributor bounties)

Everything else — GitHub sync, dashboards, leaderboards, mentor tools — is the platform's core surface.

2. Primary Use Cases

1. **Open source competitions** (e.g., a 30-day contribution sprint across 20 repos).
2. **Mentorship programs** (GSoC-style, with mentor–mentee pairing and weekly check-ins).
3. **Sustained OSS initiatives** (ongoing bounty boards, community rewards).
4. **Multi-org collaborations** (several orgs jointly running an event with pooled prize pools).

3. User Roles

Role	Description	Key Capabilities
Platform Owner (Elixpo)	Runs the PaaS	Billing, payouts, KYC, compliance, abuse handling
Org Admin	The organization hosting the initiative	Create programs, connect GitHub org/repos, invite mentors, set point rules, approve payouts
Program Admin / Maintainer	Runs a specific initiative under an org	Manage issues, labels, review submissions, adjust points
Mentor	Guides contributors	Claim mentees, review PRs, log activity, award bonus points
Contributor	Participates	Claim issues, submit PRs, track points, redeem rewards
Spectator	Public viewer	View public leaderboards and program pages

4. Core Feature Set

4.1 Program Setup

- Create a program with metadata (name, timeline, prize pool, rules).
- Choose integration mode:
 - **Full GitHub Org mode** — the entire org's repos are pulled in and tracked.
 - **Multi-repo mode** — pick specific repos across one or more orgs.
- Configure designated **issue labels** (e.g. `elixpo-easy`, `elixpo-medium`, `elixpo-hard`, `elixpo-bug`, `elixpo-feature`) that map to point values.
- Set point rules: base points per label, PR-merge bonus, first-time contributor bonus, review multipliers, streak boosts.

4.2 GitHub Integration

- GitHub App installed on the org or selected repos.
- Webhooks for: issues opened/labeled/assigned/closed, PRs opened/reviewed/merged, comments, commits.
- Two-way sync: label changes on GitHub reflect in the workspace; platform-side label suggestions can push back to GitHub.
- Automatic contributor detection via commit author + PR author.
- Anti-gaming heuristics: PR–issue linkage required, minimum diff size, mentor/admin verification for high-value awards.

4.3 Workspace Dashboard

Per-org workspace showing:

- Active programs and their health (open issues, unclaimed, stale PRs).
- **Mentor activity feed** — reviews given, mentees helped, hours logged.
- **Admin activity log** — every point adjustment, payout approval, label change (full audit trail).
- Contributor funnel — signups → first claim → first PR → merged.
- Prize pool status and payout queue.

4.4 Gamification & Leaderboard

- Global and per-program leaderboards.
- Points, levels, badges, and streaks.
- Seasonal resets (optional) so new contributors aren't perpetually behind.
- Tiebreakers: PR merge count → review count → earliest submission.
- Public shareable profile pages (`opensource.elixpo.com/u/<handle>`).

4.5 Contributor Experience

- Browse open issues across all connected repos with filters (difficulty, language, program).
- One-click claim (creates an assignment on GitHub).
- Auto-unclaim after inactivity window.
- Personal dashboard: claimed issues, open PRs, points earned, upcoming payouts.

4.6 Mentor Tools

- Mentor queue of PRs awaiting review.
- Ability to award bonus points with a reason (logged).
- Office-hour scheduling and check-in logging.
- Mentor leaderboard (recognition, not payout-gating — mentor payouts are stipend-based).

4.7 Admin Tools

- Bulk label operations across repos.
- Point rule editor with dry-run preview.
- Manual override with mandatory reason.
- Program analytics: contributor retention, issue turnaround time, cost-per-merged-PR.

4.8 Elixpo-Managed Services

These are **owned end-to-end by Elixpo** and exposed to orgs as a service — orgs do not need to bring their own auth, mail, or payout providers.

- **Account Management**
 - Unified Elixpo identity (email + GitHub OAuth + optional SSO for enterprise orgs).
 - Org membership, invitations, role assignment.
 - KYC / tax-form collection for contributors above payout thresholds.
- **Email System**
 - Transactional: invites, claim confirmations, PR-merged, points awarded, payout initiated.
 - Digest: weekly leaderboard, mentor summaries, admin health reports.
 - Deliverability handled by Elixpo (SPF/DKIM/DMARC on `opensource.elixpo.com`).
- **Payouts**
 - Prize disbursement to contributors (multi-rail: bank transfer, UPI, PayPal, crypto where legal).
 - Mentor stipends on a recurring schedule.
 - Bounty payouts triggered by admin approval + PR-merged event.
 - Full ledger, invoicing to the org, and tax reporting.

5. High-Level Architecture

flowchart TB

GH[GitHub Apps API]

FE[Web Frontend
Next.js]

DASH[Mentor / Admin
Dashboards]

GW[API Gateway
auth · rate-limit · routing]

PROG[Programs Service]

PTS[Points & Gamification Service]

SYNC[GitHub Sync Worker]

DB[(Postgres
Primary DB)]

BUS[[Event Bus
Redis / NATS]]

AUTH[Auth / Identity
Elixpo]

MAIL[Email Service
Elixpo]

PAY[Payouts Service
Elixpo]

GH -- webhooks / REST / GraphQL --> GW

FE --> GW

DASH --> GW

GW --> PROG

GW --> PTS

GW --> SYNC

PROG --> DB

PTS --> DB

SYNC --> DB

PROG --> BUS

PTS --> BUS

SYNC --> BUS

BUS --> AUTH

BUS --> MAIL

BUS --> PAY

GW -. -> AUTH

classDef elixpo fill:#4f46e5,stroke:#312e81,color:#fff;

class AUTH,MAIL,PAY elixpo;

6. Suggested Tech Stack

- **Frontend:** Next.js + Tailwind + shadcn/ui
- **Backend:** Node.js (NestJS) or Python (FastAPI) microservices
- **DB:** PostgreSQL (primary), Redis (cache + queues), ClickHouse (analytics/leaderboards at scale)
- **Event bus:** NATS or Redis Streams
- **GitHub:** GitHub App (not OAuth app) for org-level install
- **Auth:** Elixpo Identity (JWT + refresh), GitHub OAuth linked
- **Email:** Elixpo mail service (backed by SES/Postmark), templated via MJML
- **Payouts:** Elixpo payout service abstracting Razorpay / Stripe Connect / Wise
- **Infra:** Docker + Kubernetes (or Fly.io/Railway for early stage), Terraform-managed
- **Observability:** OpenTelemetry, Grafana, Sentry

7. Data Model (Core Entities)

- **Organization** — the hosting entity
- **Program** — a competition/initiative under an org
- **Repository** — linked GitHub repos (belong to a program)
- **LabelRule** — label → point mapping per program
- **User** — Elixpo identity
- **Membership** — user ↔ org ↔ role

- **Contribution** — issue claims, PRs, reviews
- **PointsLedger** — append-only, every award/deduction with reason + actor
- **Payout** — status machine: pending → approved → processing → paid/failed
- **AuditLog** — every admin/mentor action

8. Gamification Rules (Default Template)

Event	Points
elixpo-easy merged	10
elixpo-medium merged	25
elixpo-hard merged	60
First PR of the program	+15 bonus
PR review (approved by maintainer)	5
7-day contribution streak	+20
Mentor-awarded bonus	up to 25 (with reason)
PR closed without merge	0
Spam / rule violation	negative + review

All values overridable per program.

9. Anti-Abuse & Fair Play

- Require PR to reference a claimed issue.
- Minimum diff / commit heuristics + maintainer approval before points settle.
- Cool-down between claims.
- Rate-limit label edits from non-admins.
- Shadow flags for suspicious accounts (new account + high volume + trivial PRs).
- Human-in-the-loop for payouts above configurable threshold.

10. Rollout Roadmap

- Phase 0 — Foundations** Domain, identity (Elixpo auth), GitHub App skeleton, org onboarding, basic dashboard.
- Phase 1 — Core Program Flow** Program creation, repo linking (both modes), label rules, issue claim flow, points ledger, contributor dashboard.
- Phase 2 — Gamification & Leaderboards** Leaderboards, badges, streaks, public profile pages, email digests.
- Phase 3 — Mentor & Admin Suite** Mentor queue, admin activity log, audit trail, bulk operations, analytics.
- Phase 4 — Payouts & Compliance** Elixpo payout integration, KYC flow, tax forms, invoicing to orgs. Support for orgs bringing their own payout provider via integration adapter.
- Phase 5 — Polish & Scale** Anti-abuse hardening, ClickHouse-backed analytics, enterprise SSO, public API, webhooks for orgs.

11. Success Metrics

- Time-to-first-program (target: < 15 min from signup).
- Contributor activation rate (signup → first merged PR).
- Program NPS from org admins.
- Payout success rate & median time-to-payout.
- Mentor retention across programs.

12. Resolved Decisions

-
- **Source control support (v1):** GitHub **and** GitLab are supported. Bitbucket is **not** in v1 — deferred to a later phase based on demand.
 - **Payout providers:** Elixpo payouts are the default managed rail, but orgs **can bring their own payout provider** — they are responsible for the integration against the platform's payout adapter interface. Elixpo provides the contract, the org provides the implementation.
 - **Cross-program contributor identity:** Leaderboards are **siloed per program**. Each program runs its own leaderboard and prize distribution independently — no global cross-program ranking in v1.
 - **Terms of Service & Privacy Policy:** Deferred. Will be drafted before public launch, not blocking build.
-

Prepared for the Elixpo team — opensource.elixpo.com platform planning.